

Topics: 14.1 Functions of Several Variables; 14.2 Limits and Continuity; 14.3 Partial Derivatives (Part I)

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1. A scuba air tank is built from a right circular cylinder of radius x and height y , with two hemispherical end caps of radius x attached (one on each end of the cylinder).

- (a) Express the volume V of the tank as a function of x and y .

The volume of the tank equals the volume of the cylinder plus the volume of the two hemispheres:

$$V(x, y) = \pi x^2 y + 2 \cdot \frac{2}{3} \pi x^3 = \pi x^2 y + \frac{4}{3} \pi x^3.$$

- (b) Evaluate $V(3, 2)$ and explain the meaning of this value.

Evaluating $V(x, y)$ at $(3, 2)$, we get that

$$V(3, 2) = \pi(3)^2(2) + \frac{4}{3}\pi(3)^3 = 18\pi + 36\pi = 54\pi.$$

The volume of such a tank with radius $x = 3$ and cylindrical height $y = 2$ is 54π (≈ 170) units³.

2. Consider the function $F(x, y, z) = \sqrt{y} - \sqrt{x - 2z}$.

- (a) Evaluate $F(3, 4, 1)$.

Substitute $x = 3$, $y = 4$, and $z = 1$:

$$F(3, 4, 1) = \sqrt{4} - \sqrt{3 - 2(1)} = \sqrt{4} - \sqrt{1} = 2 - 1 = 1.$$

- (b) Find the domain of F .

For $F(x, y, z)$ to be defined, both expressions under the square roots must be nonnegative:

$$y \geq 0 \quad \text{and} \quad x - 2z \geq 0.$$

Therefore the domain is

$$\{(x, y, z) \in \mathbb{R}^3 : y \geq 0 \text{ and } x - 2z \geq 0\}.$$

3. Find and sketch the domain of the function.

- (a) $f(x, y) = \frac{\ln(2 - x)}{1 - x^2 - y^2}$

For $f(x, y)$ to be defined, the logarithm must have a positive argument. Thus,

$$2 - x > 0 \implies x < 2.$$

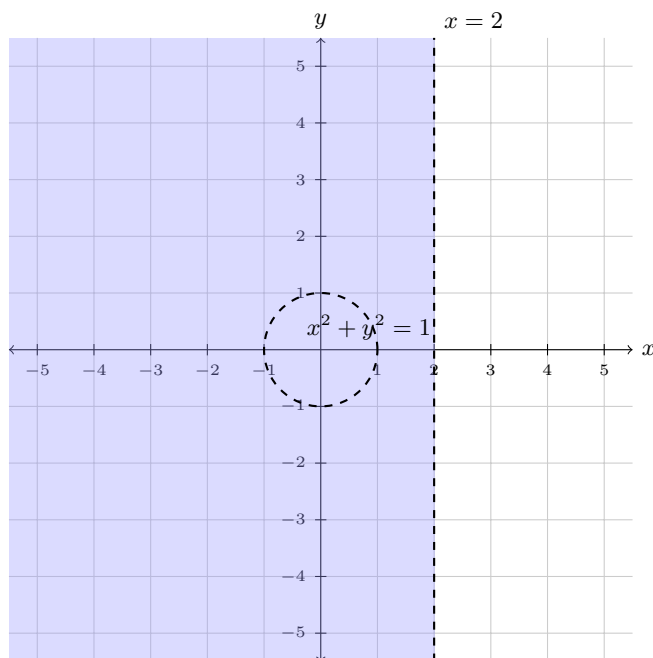
Also, the denominator cannot be zero, so we must require:

$$1 - x^2 - y^2 \neq 0 \implies x^2 + y^2 \neq 1.$$

Therefore the domain is

$$D = \{(x, y) \in \mathbb{R}^2 : x < 2 \text{ and } x^2 + y^2 \neq 1\}.$$

Geometrically, this is the half-plane to the left of the line $x = 2$, with the circle $x^2 + y^2 = 1$ removed.



$$(b) \ g(x, y) = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x^2 + y^2 - 4}}$$

The function g is defined only when all radicals are real and the denominator is nonzero.

Since $\sqrt{x}$ and $\sqrt{y}$ appear in the numerator, we must have

$$x \geq 0 \quad \text{and} \quad y \geq 0.$$

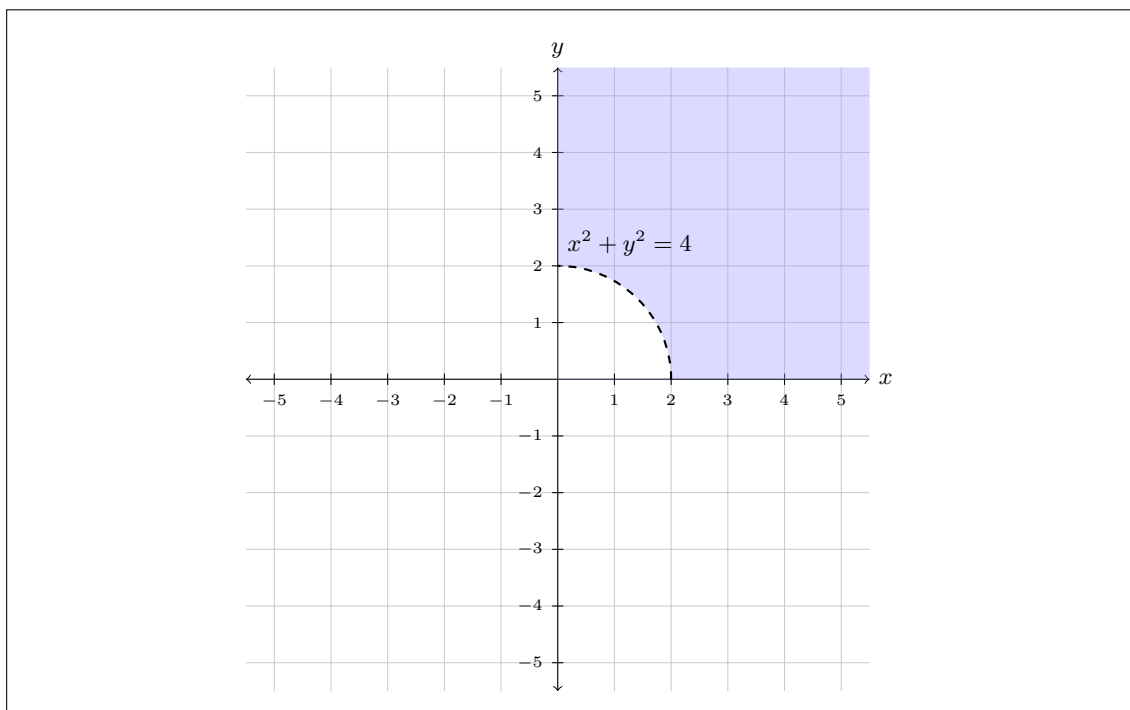
Also, the expression under the square root in the denominator must be positive (not merely non-negative, because it is in the denominator):

$$x^2 + y^2 - 4 > 0 \implies x^2 + y^2 > 4.$$

Therefore the domain of g is

$$D = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0 \text{ and } x^2 + y^2 > 4\}.$$

Geometrically, this is the part of the closed first quadrant lying outside the circle $x^2 + y^2 = 4$.



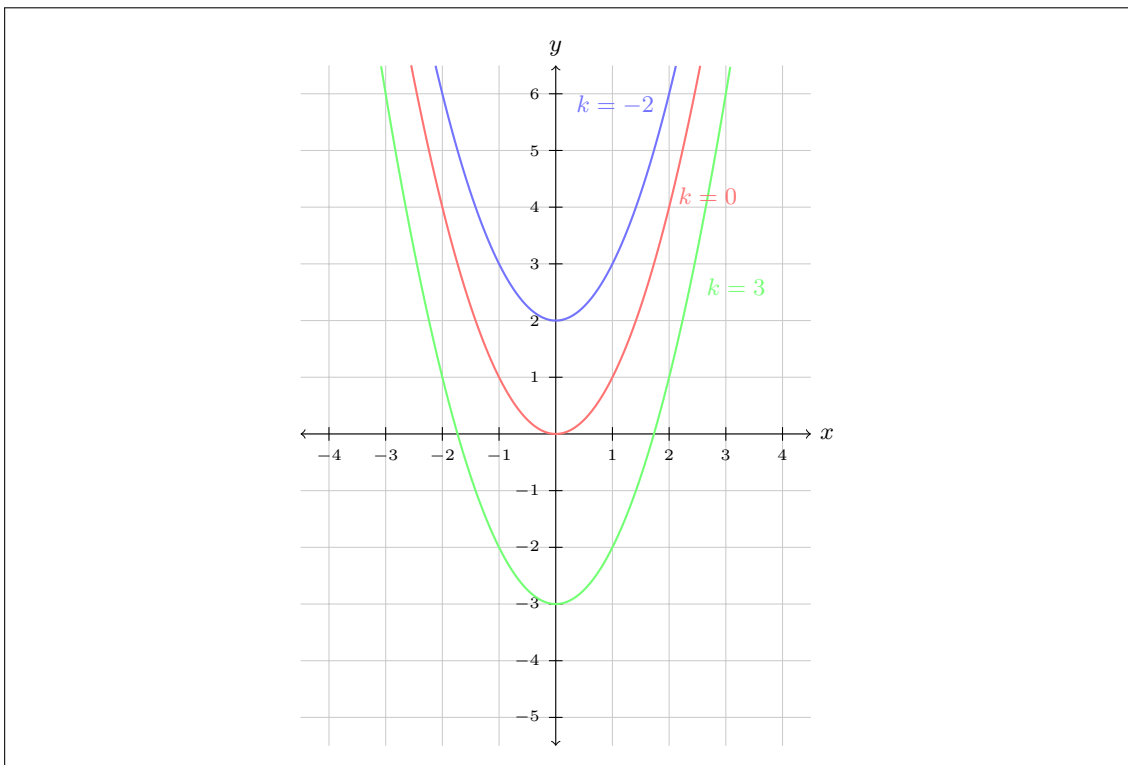
4. Sketch the level curves of the function for the given values of k .

(a) $f(x, y) = x^2 - y$, for $k = -2, 0, 3$.

A level curve $f(x, y) = k$ satisfies

$$x^2 - y = k \implies y = x^2 - k,$$

so the level curves are parabolas opening upward.

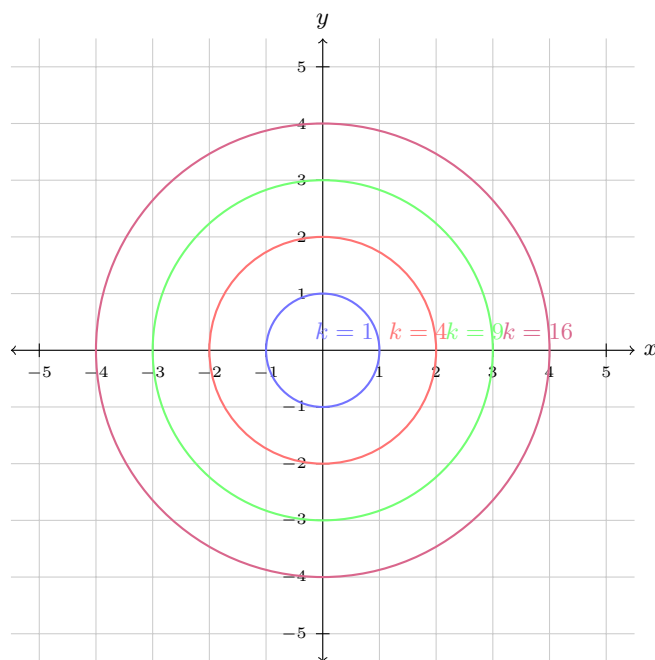


(b) $g(x, y) = x^2 + y^2$, for $k = 1, 4, 9, 16$.

A level curve $g(x, y) = k$ satisfies

$$x^2 + y^2 = k,$$

which is a circle centered at the origin with radius $\sqrt{k}$. Thus the requested level curves are circles of radii 1, 2, 3, and 4 centered at the origin.



5. Evaluate the limit.

$$(a) \lim_{(x,y) \rightarrow (1,1)} \frac{x^2y^3 - x^3y^2}{x^2 - y^2}$$

We evaluate the limit by factoring:

$$\lim_{(x,y) \rightarrow (1,1)} \frac{x^2y^3 - x^3y^2}{x^2 - y^2} = \lim_{(x,y) \rightarrow (1,1)} \frac{x^2y^2(y - x)}{(x - y)(x + y)} = \lim_{(x,y) \rightarrow (1,1)} -\frac{x^2y^2}{x + y} = -\frac{1}{2}.$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1}$$

We evaluate the limit by using the conjugate.

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1} &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1} \cdot \frac{\sqrt{x^2 + y^2 + 1} + 1}{\sqrt{x^2 + y^2 + 1} + 1} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)(\sqrt{x^2 + y^2 + 1} + 1)}{(x^2 + y^2 + 1) - 1} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)(\sqrt{x^2 + y^2 + 1} + 1)}{x^2 + y^2} \\ &= \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2 + 1} + 1 \\ &= \sqrt{0^2 + 0^2 + 1} + 1 \\ &= 2. \end{aligned}$$

$$(c) \lim_{(x,y,z) \rightarrow (1,-1,1)} \frac{2xz + 3x + 2yz + 3y}{x + y}$$

We evaluate the limit by factoring:

$$\begin{aligned} \lim_{(x,y,z) \rightarrow (1,-1,1)} \frac{2xz + 3x + 2yz + 3y}{x + y} &= \lim_{(x,y,z) \rightarrow (1,-1,1)} \frac{(x + y)(2z + 3)}{x + y} \\ &= \lim_{(x,y,z) \rightarrow (1,-1,1)} 2z + 3 \\ &= 2(1) + 3 \\ &= 5. \end{aligned}$$

6. Show that the limit does not exist.

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{3x - 2y}{4x^2 - y^2}$$

Let's approach (0,0) along the line $y = x$:

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=x}} \frac{3x - 2y}{4x^2 - y^2} = \lim_{x \rightarrow 0} \frac{3x - 2x}{4x^2 - x^2} = \lim_{x \rightarrow 0} \frac{x}{3x^2} = \lim_{x \rightarrow 0} \frac{1}{3x}.$$

This one-variable limit does not exist (it is unbounded and approaches $+\infty$ from the right and $-\infty$ from the left). Therefore the two-variable limit does not exist as it does not exist along a path approaching $(0, 0)$.

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2 + 3y^2}$

Let's approach $(0, 0)$ along the line $y = ax$, where a is a constant. Then

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=ax}} \frac{2xy}{x^2 + 3y^2} = \lim_{x \rightarrow 0} \frac{2x(ax)}{x^2 + 3(ax)^2} = \lim_{x \rightarrow 0} \frac{2ax^2}{(1 + 3a^2)x^2} = \frac{2a}{1 + 3a^2}.$$

Because the resulting value depends on a , different paths give different limits. For instance,

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ y=x}} \frac{2xy}{x^2 + 3y^2} = \frac{2}{1 + 3} = \frac{1}{2}, \quad \lim_{\substack{(x,y) \rightarrow (0,0) \\ y=2x}} \frac{2xy}{x^2 + 3y^2} = \frac{4}{1 + 12} = \frac{4}{13}.$$

Since $\frac{1}{2} \neq \frac{4}{13}$, the limit does not exist.

7. Determine the set of points at which the function $f(x, y) = \ln(1 - x - y)$ is continuous.

The logarithm function is continuous wherever its argument is positive. Thus, we require

$$1 - x - y > 0.$$

Therefore, f is continuous on the set

$$\{(x, y) \in \mathbb{R}^2 : 1 - x - y > 0\}.$$

Geometrically, this is the open half-plane strictly below the line $y = 1 - x$.

8. Find all first order partial derivatives of the function.

(a) $f(x, y) = (x^2y - y^3)^5$

$$f_x(x, y) = 5(x^2y - y^3)^4(2xy), \quad f_y(x, y) = 5(x^2y - y^3)^4(x^2 - 3y^2)$$

(b) $g(x, y, z) = 4x^3y^2 - e^zy^4 + \frac{z^3}{x^2} + 4y - x^5$

$$g_x(x, y, z) = 12x^2y^2 - \frac{2z^3}{x^3} - 5x^4, \quad g_y(x, y, z) = 8x^3y - 4e^zy^3 + 4, \quad g_z(x, y, z) = -e^zy^4 + \frac{3z^2}{x^2}$$

(c) $h(x, y, z) = \frac{x \sin y}{z^2}$

$$h_x(x, y, z) = \frac{\sin y}{z^2}, \quad h_y(x, y, z) = \frac{x \cos y}{z^2}, \quad h_z(x, y, z) = -\frac{2x \sin y}{z^3}.$$

9. Consider the function $f(x, y) = y \arcsin(xy)$.

(a) Find the partial derivatives f_x and f_y .

Taking the partial derivative with respect to x ,

$$f_x(x, y) = y \cdot \frac{1}{\sqrt{1 - (xy)^2}} \cdot y = \frac{y^2}{\sqrt{1 - x^2y^2}}.$$

Taking the partial derivative with respect to y ,

$$f_y(x, y) = \arcsin(xy) + y \cdot \frac{1}{\sqrt{1 - (xy)^2}} \cdot x = \arcsin(xy) + \frac{xy}{\sqrt{1 - x^2y^2}}.$$

(b) Evaluate $f_x(1, \frac{1}{2})$. Interpret this value as a slope.

Evaluating f_x at the point $(1, \frac{1}{2})$,

$$f_x\left(1, \frac{1}{2}\right) = \frac{(1/2)^2}{\sqrt{1 - (1/2)^2}} = \frac{1/4}{\sqrt{3}/2} = \frac{\sqrt{3}}{6}.$$

The graph of f is the surface $z = f(x, y)$. The plane $y = \frac{1}{2}$ intersects the surface in a curve, and the slope of the tangent line to this curve at the point $(1, \frac{1}{2}, f(1, \frac{1}{2})) = (1, \frac{1}{2}, \frac{\pi}{12})$ is $f_x(1, \frac{1}{2}) = \frac{\sqrt{3}}{6}$.

(c) Evaluate $f_y(1, \frac{1}{2})$. Interpret this value as a slope.

Evaluating f_y at the point $(1, \frac{1}{2})$,

$$f_y\left(1, \frac{1}{2}\right) = \arcsin\left(\frac{1}{2}\right) + \frac{1/2}{\sqrt{3}/2} = \frac{\pi}{6} + \frac{\sqrt{3}}{3}.$$

Similarly, the plane $x = 1$ intersects the surface in a curve, and the slope of the tangent line to this curve at the point $(1, \frac{1}{2}, \frac{\pi}{12})$ is $f_y(1, \frac{1}{2}) = \frac{\pi}{6} + \frac{\sqrt{3}}{3}$.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Functions of Two and Three Variables. A function of two variables is a rule that assigns to each point (x, y) in a subset of $\mathbb{R}^2$ a real number. A function of three variables is a rule that assigns to each point (x, y, z) in a subset of $\mathbb{R}^3$ a real number.

Unless otherwise specified, by the **domain** of $f(x, y)$, we mean the set of all points (x, y) for which the expression for f produces a real number (similarly for functions of three variables).

Level Curves and Level Surfaces. The **level curve** of $f(x, y)$ corresponding to the value k is the curve given by

$$f(x, y) = k.$$

For functions of three variables, the analogous objects are **level surfaces**, given by

$$g(x, y, z) = k.$$

Limits. If $f(x, y)$ can be made arbitrarily close to L by taking (x, y) sufficiently close to (a, b) (but not equal to (a, b)), then we write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L.$$

On the other hand, if $f(x, y)$ does not approach a single value as (x, y) approaches (a, b) , then the above limit does not exist. Analogous definitions apply for functions of three variables.

Partial Derivatives. Let f be a function of two variables, and write $z = f(x, y)$. The partial derivative of f with respect to x is denoted by f_x or $\frac{\partial f}{\partial x}$ and represents the rate of change of z with respect to x when y is fixed. To find f_x , we regard y as a constant and differentiate f with respect to x . Similarly, the partial derivative of f with respect to y is denoted by f_y or $\frac{\partial f}{\partial y}$ and represents the rate of change of z with respect to y when x is fixed. Partial derivatives are also defined similarly for functions of three or more variables.

Suggested Textbook Problems

Section 14.1: 1-56, 61-72

Section 14.2: 1-50

Section 14.3: 1-74